WORKED SOLUTION

STEP III 1991, Question 8

Pure • Reduction Formulae and Sums

STEP III 1991, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q8

The key idea

Part (i) is a reduction formula, so integration by parts is the natural first move. The leftover integrand contains $\sin x \sin kx$, and the compound-angle formula for $\cos(k-1)x$ turns it back into integrals of the same family. Part (ii) is the real part of a binomial expansion of $(1+\mathrm{e}^{2\mathrm{i}\theta})^m$. Part (iii) then comes from integrating (ii) and unwinding the recurrence from (i) by telescoping.

Solution

Part (i)

Take $k \geq 1$ and integrate by parts with $u = \cos^k x$ and $\mathrm{d}v = \cos kx\,\mathrm{d}x$, so that $v = \frac{1}{k}\sin kx$ and $\mathrm{d}u = -k\cos^{k-1}x\sin x\,\mathrm{d}x$:

$$\begin{aligned} I_k &= \left[\tfrac{1}{k}\cos^k x \sin kx\right]_0^\theta + \int_0^\theta \cos^{k-1}x\,\sin x\,\sin kx\,\mathrm{d}x \\ &= \tfrac{1}{k}\cos^k\theta\sin k\theta + \int_0^\theta \cos^{k-1}x\,\sin x\,\sin kx\,\mathrm{d}x, \end{aligned}$$

since the bracket vanishes at $x = 0$.

Ask yourself How can $\sin x \sin kx$ be rewritten in terms of cosines of multiples of x?

Writing $(k-1)x = kx - x$, the compound-angle formula gives

$$\cos(k-1)x = \cos kx\cos x + \sin kx\sin x,$$

so

$$\sin x\sin kx = \cos(k-1)x - \cos x\cos kx.$$

Therefore

$$\begin{aligned}\int_0^\theta \cos^{k-1} x \, \sin x \, \sin kx \, \mathrm{d}x &= \int_0^\theta \cos^{k-1} x \cos(k-1)x \, \mathrm{d}x \\ &\quad - \int_0^\theta \cos^k x \cos kx \, \mathrm{d}x \\ &= I_{k-1} - I_k.\end{aligned}$$

Substituting back and multiplying by k,

$$kI_k = \cos^k \theta \sin k\theta + kI_{k-1} - kI_k,$$

that is,

$$2kI_k = kI_{k-1} + \cos^k \theta \sin k\theta,$$

as required.

Part (ii)

Let $z = \mathrm{e}^{2\mathrm{i}\theta}$. By the binomial theorem,

$$\begin{aligned}(1+z)^m &= \sum_{r=0}^{m} \binom{m}{r} z^r \\ &= \sum_{r=0}^{m} \binom{m}{r} (\cos 2r\theta + \mathrm{i} \sin 2r\theta).\end{aligned}$$

Now factor out the "half angle":

$$1 + \mathrm{e}^{2\mathrm{i}\theta} = \mathrm{e}^{\mathrm{i}\theta}(\mathrm{e}^{-\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}\theta}) = 2\cos\theta \, \mathrm{e}^{\mathrm{i}\theta},$$

so

$$(1+z)^m = 2^m \cos^m \theta \, \mathrm{e}^{\mathrm{i}m\theta} = 2^m \cos^m \theta \, (\cos m\theta + \mathrm{i} \sin m\theta).$$

Since $2^m \cos^m \theta$ is real, equating real parts of the two expressions gives

$$\sum_{r=0}^{m} \binom{m}{r} \cos 2r\theta = 2^m \cos^m \theta \cos m\theta,$$

which is the required identity (the terms $r = 0$ and $r = m$ are 1 and $\cos 2m\theta$).

Part (iii)

Call the first expression S. The right-hand side of (ii) is 2^m times the integrand of I_m, so we integrate (ii) from 0 to θ. The $r = 0$ term integrates to θ, and for $r \geq 1$, $\int_0^\theta \cos 2rx \, \mathrm{d}x = \frac{\sin 2r\theta}{2r}$. Hence

$$\theta + \sum_{r=1}^{m} \binom{m}{r} \frac{\sin 2r\theta}{2r} = 2^m I_m, \qquad \text{so} \qquad S = 2^m I_m - \theta.$$

Now we evaluate $2^m I_m$ using (i). Dividing the recurrence by $2k$ and multiplying by 2^k gives, for $k \geq 1$,

$$2^k I_k = 2^{k-1} I_{k-1} + \frac{2^{k-1}}{k} \cos^k \theta \sin k\theta.$$

Ask yourself Why multiply by 2^k?

Written as $I_k = \frac{1}{2} I_{k-1} + \ldots$, the recurrence has a factor $\frac{1}{2}$ at each step. Using $J_k = 2^k I_k$ removes it: $J_k - J_{k-1}$ is then a simple explicit term, and the sum telescopes. Adding the equations for $k = 1, 2, \ldots, m$, the intermediate terms cancel:

$$2^m I_m = 2^0 I_0 + \sum_{k=1}^{m} \frac{2^{k-1}}{k} \cos^k \theta \sin k\theta.$$

Finally $I_0 = \int_0^\theta 1 \, \mathrm{d}x = \theta$, so

$$\begin{aligned} S = 2^m I_m - \theta &= \sum_{k=1}^{m} \frac{1}{k} 2^{k-1} \cos^k \theta \sin k\theta \\ &= \cos\theta \sin\theta + \cos^2\theta \sin 2\theta + \cdots + \frac{1}{m} 2^{m-1} \cos^m \theta \sin m\theta, \end{aligned}$$

which is the required result.

Checks

- $m = 1$: the claim is $\frac{1}{2} \sin 2\theta = \cos\theta \sin\theta$, which is the double-angle formula.
- $m = 2$: the left side is $\sin 2\theta + \frac{1}{4} \sin 4\theta$, and the right side is $\cos\theta \sin\theta + \cos^2\theta \sin 2\theta = \frac{1}{2} \sin 2\theta + \frac{1}{2} \sin 2\theta (1 + \cos 2\theta)$, which is the same, since $\sin 2\theta \cos 2\theta = \frac{1}{2} \sin 4\theta$.
- Numerically, with $m = 5$ and $\theta = 0.7$ both expressions equal 1.51963 (to 5 decimal places). The reduction formula also checks against Simpson's-rule values of I_k for random k and θ.

Common slips

- Sign errors in the integration by parts: $\frac{\mathrm{d}}{\mathrm{d}x} \cos^k x = -k \cos^{k-1} x \sin x$, and that minus sign cancels the minus in front of $\int v \, \mathrm{d}u$.
- Forgetting the $r = 0$ term in (ii). It integrates to θ, and it cancels the $I_0 = \theta$ at the bottom of the telescoping sum.

Beyond STEP

Writing $\cos^m\theta\cos m\theta$ as a combination of $\cos 2r\theta$ amounts to computing a Fourier series. Here it is a finite one, and integrating term by term is exactly how Fourier series are handled in a first-year methods course. Reduction formulae of the kind in (i) are also the standard tool for integrals of the form $\int \cos^n x\,\mathrm{d}x$ and for the Wallis product. Needham's *Visual Complex Analysis* gives a geometric picture of the trick $1+\mathrm{e}^{2\mathrm{i}\theta}=2\cos\theta\,\mathrm{e}^{\mathrm{i}\theta}$ used in (ii).

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